G_{Ep}/G_{Mp} with an 11 GeV Electron Beam

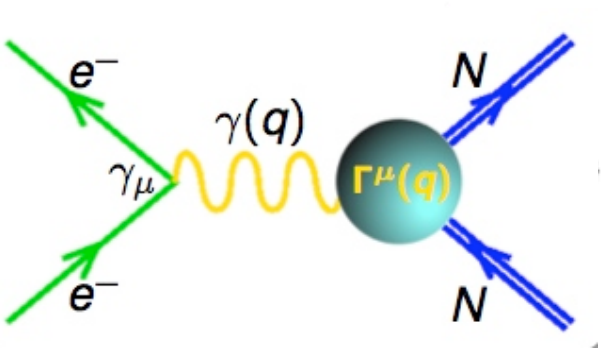
E.J. Brash, M.K. Jones,
C.F. Perdrisat, V. Punjabi

For the Gep-IV Collaboration

Elastic EM Form Factors

- Fundamental to our understanding of nucleon structure
  - At low Q^2 (larger distance), they shed information on the pion cloud.
  - At large Q^2 (shorter distance), they contain information on the quark and gluon structure
- At JLab energies, we gain knowledge about the non-perturbative structure of the nucleon
- Together with other observables (in DIS, SIDIS, DVCS, RCS, etc.), we develop a greater understanding of the nucleon's rich internal structure
- Convenient "meeting place" between theory and experiment

Elastic Form Factors



Nucleon current operator (Dirac & Pauli)

$$\Gamma^{\mu}(q) = \gamma^{\mu} F_1(q^2) + \frac{i}{2M_N}\sigma^{\mu\nu} q_\nu F_2(q^2)$$

$F_1(Q^2)$: non spin-flip Dirac form factor

$F_2(Q^2)$: spin-flip Pauli form factor

$$F_1^p(0) = 1, \quad F_1^n(0) = 0$$
$$F_2^p(0) = \kappa_p, \quad F_2^n(0) = \kappa_n$$

Electric form factor: $G_E(Q^2) = F_1(Q^2) - \tau F_2(Q^2)$

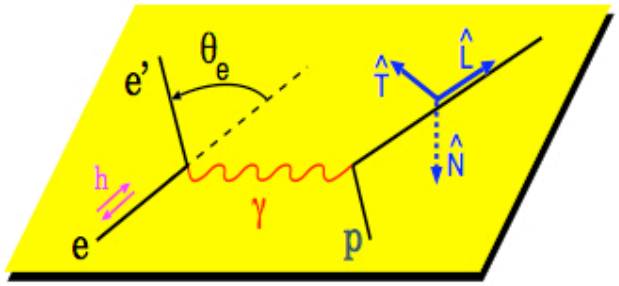
Magnetic form factor: $G_M(Q^2) = F_1(Q^2) + F_2(Q^2)$

where $\tau = \frac{Q^2}{4M_p^2}$

- Details of nucleon substructure are in the Q^2 evolution of $F_1(Q^2)$ and $F_2(Q^2)$.
- Provides a convenient "meeting place" between theory and experiment.

Recoil Polarization Method

- Elastic $\vec{e}p \to e\vec{p}$ (A. I. Akhiezer and M. P. Rekalo, Sov. J. Part. Nuc. **3**, (1974) 277; and Arnold, Carlson and Gross, Phys. Rev. **C23** (1981) 363):



$$P_n = 0$$

$$I_0 P_t = -2\sqrt{\tau(1+\tau)}\, G_{Ep} G_{Mp} \tan\tfrac{\theta_e}{2}$$

$$I_0 P_\ell = \tfrac{1}{M_p}(E_e + E_{e'})\sqrt{\tau(1+\tau)}\, G_{Mp}^2 \tan^2\tfrac{\theta_e}{2}$$

$$I_0 = G_{Ep}^2 + \tau G_{Mp}^2 \left[1 + 2(1+\tau)\tan^2\tfrac{\theta_e}{2}\right]$$

Relative Measurement:

$$\frac{G_{Ep}}{G_{Mp}} = -\frac{P_t}{P_\ell}\frac{(E_e + E_{e'})}{2M_p}\tan\left(\frac{\theta_e}{2}\right)$$

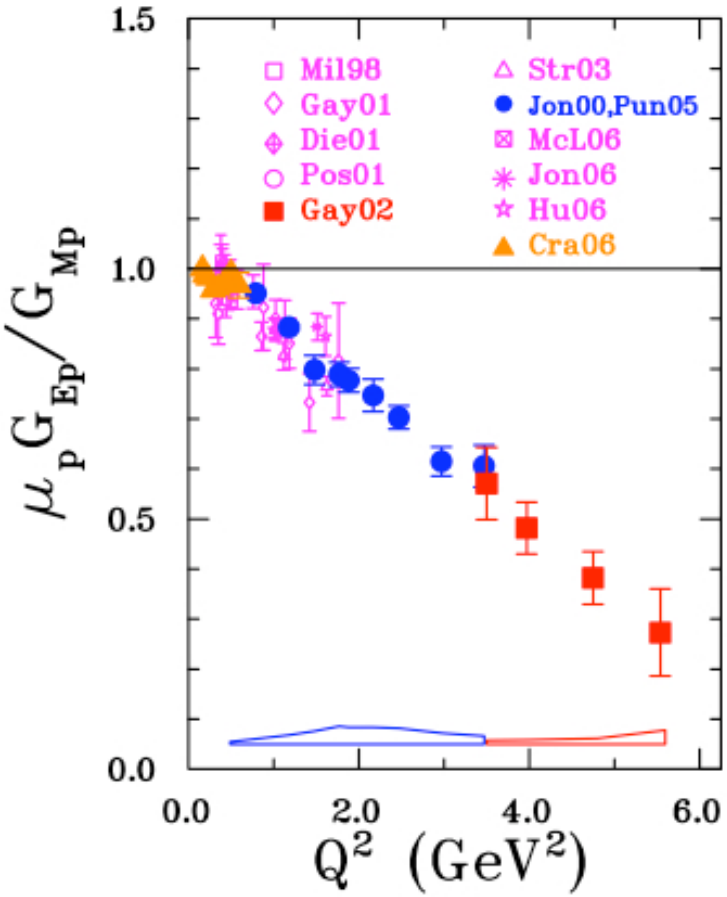
Recoil Polarization Measurements

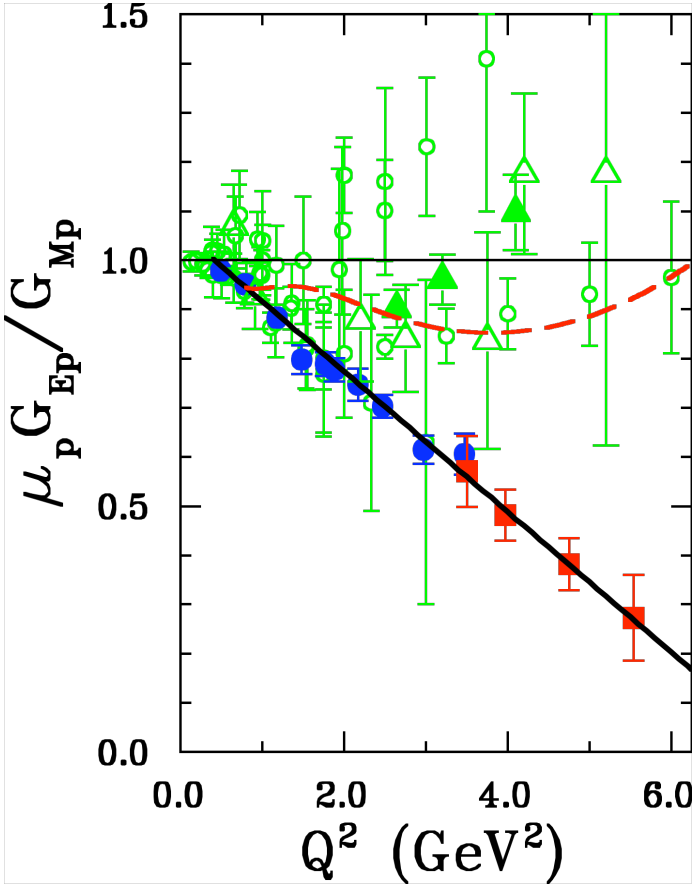
Famous Papers (250-499 citations)

M.K. Jones et al., Phys.Rev.Lett., 84 (2000) 1398 (G_{Ep}-I)

O. Gayou et al., Phys.Rev.Lett., 88 (2002) 092301. (G_{Ep}-II)

(552 total citations, 2 Ph.D.'s, 3 Masters, ~8 UG theses, several tenure track faculty and staff appointments)





G_{Ep}-III Experiment at JLab

- The ratio G_{Ep}/G_{Mp} was measured with the recoil polarization technique at Q^2 of 5.2, 6.8 and 8.54 GeV^2 in Hall C at JLab, between October 2007 and June 2008. Also G_{Ep}/G_{Mp} at a constant Q^2 of 2.49 GeV^2 for 3 kinematics (3 values of ε).

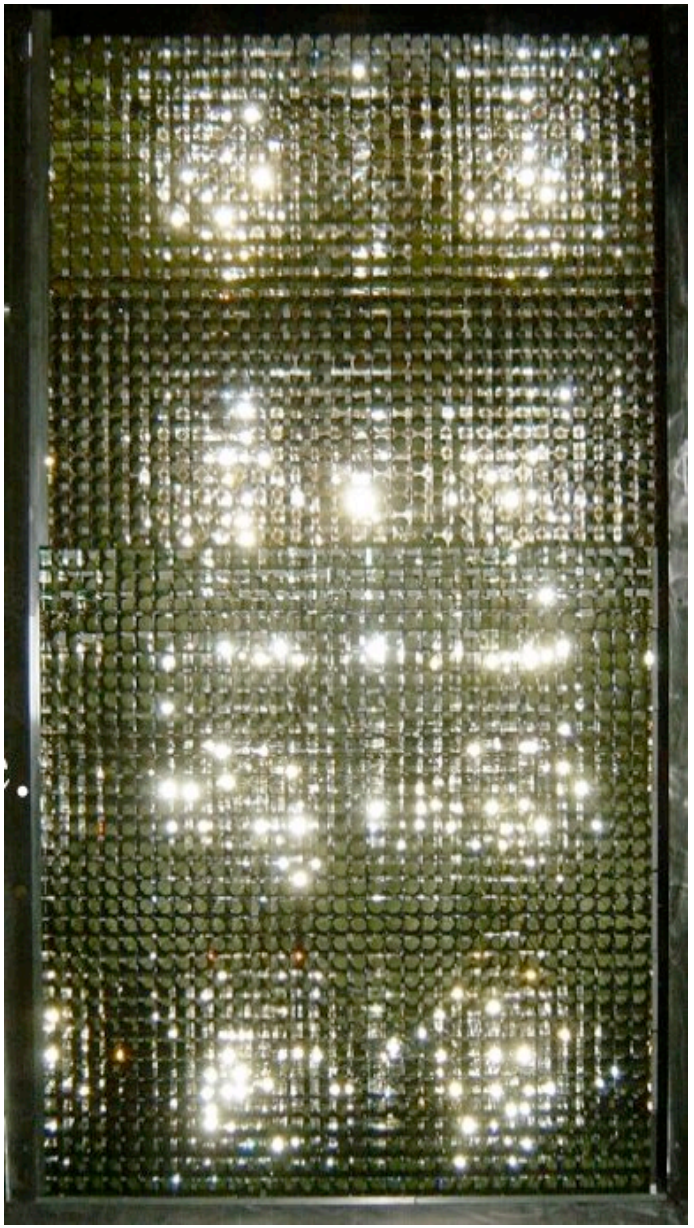
- These experiments used the high momentum spectrometer (HMS) to detect proton; a new double focal plane polarimeter (FPP) in the focal plane of the HMS measured the polarization of the recoil proton.

- A large area Electromagnetic Calorimeter (BigCal) assembled for this experiment was used to detect the elastically scattered electrons in coincidence with the protons.

BigCal in Hall C



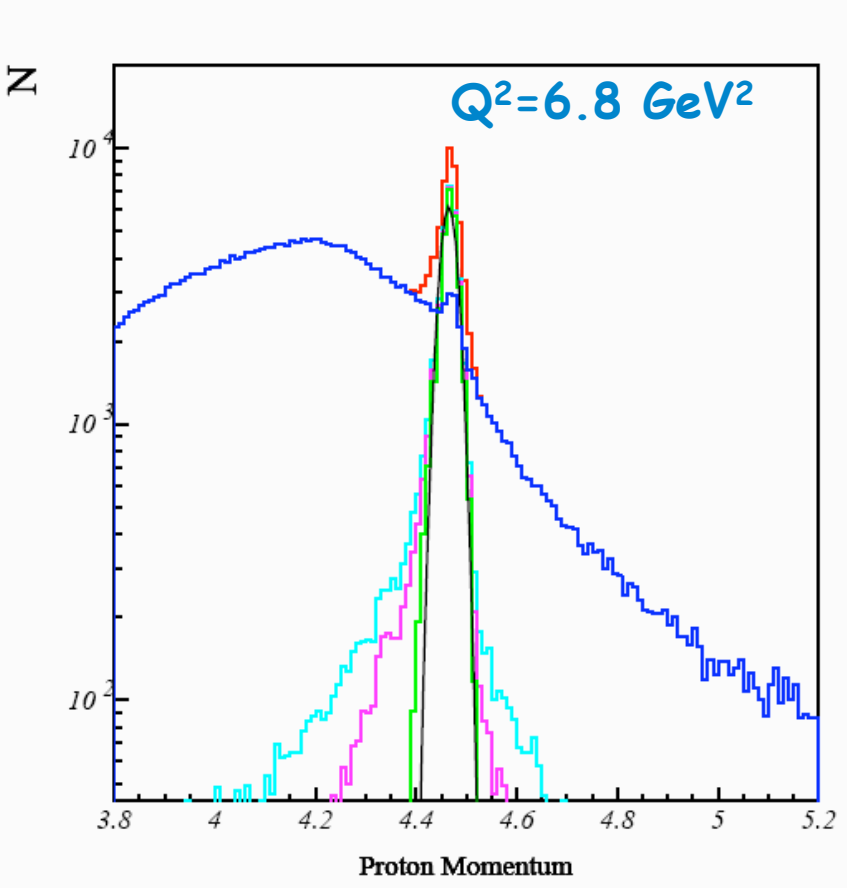
BigCal glass

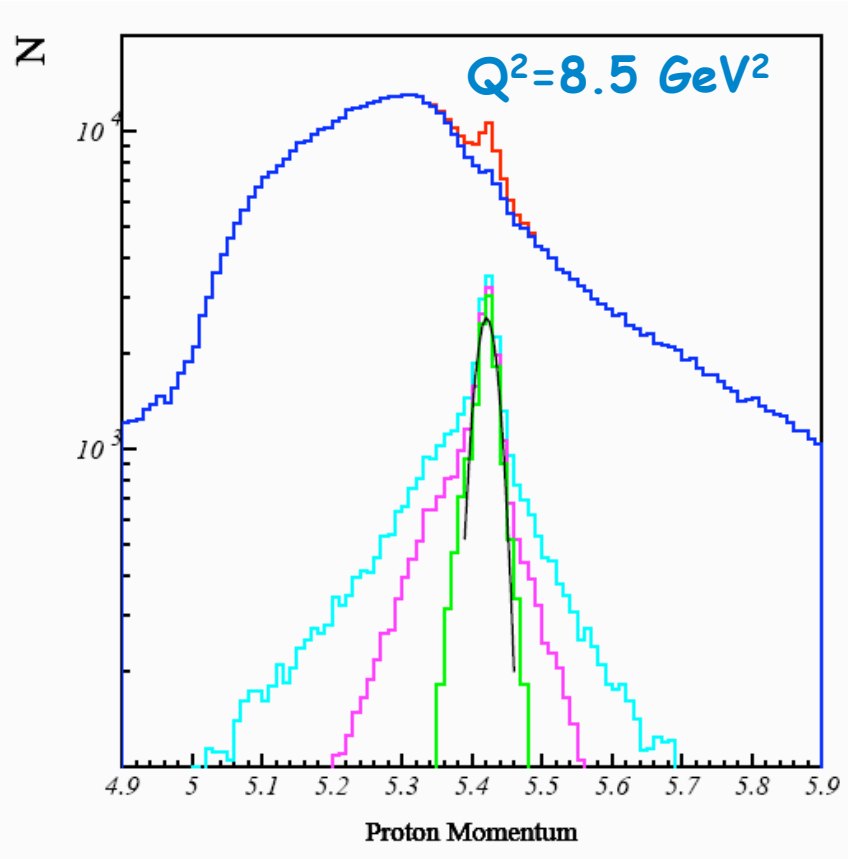


Double Focal Plane Polarimeter

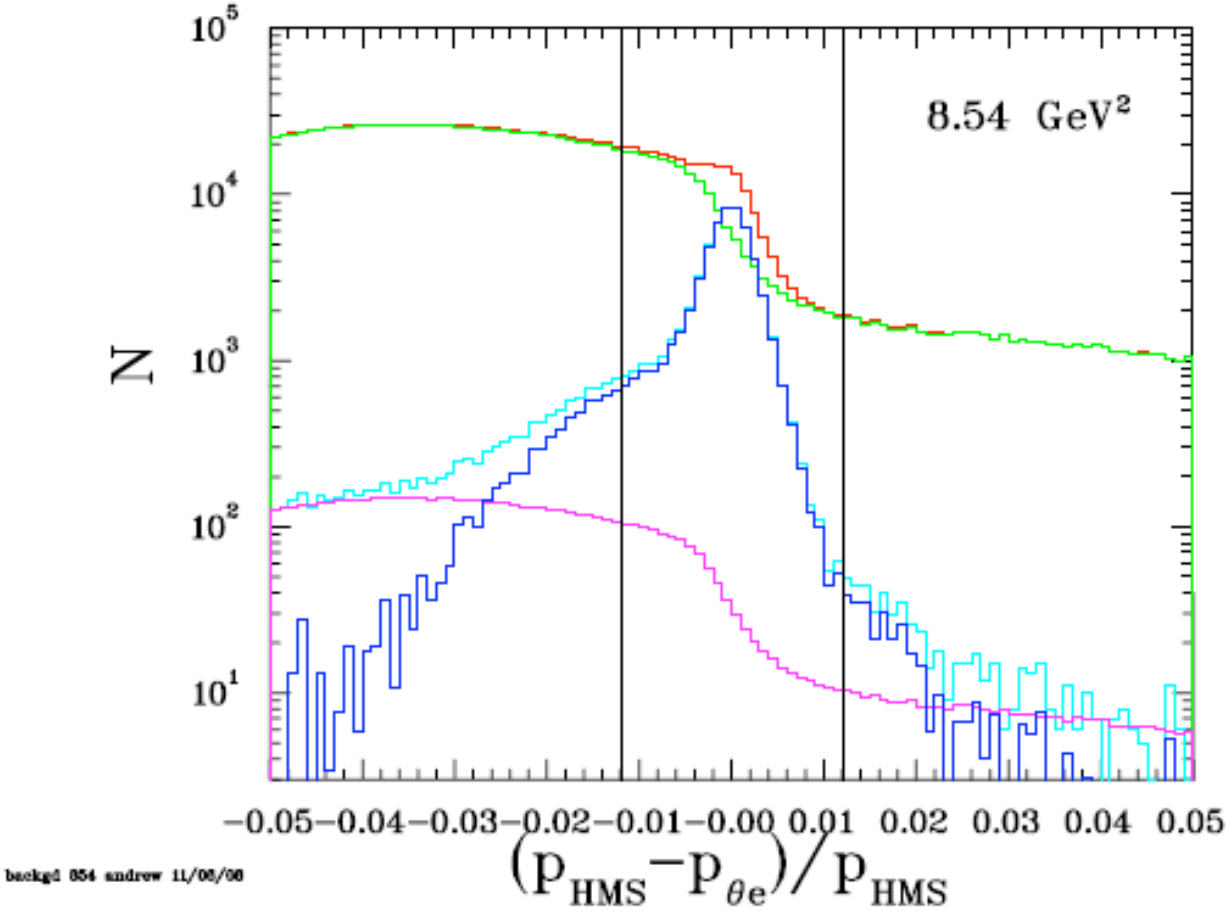


Identification of Elastic Events: Proton Momentum Spectrum





Red : all events, Cyan: with $\delta - \theta$ cut, Magenta: requiring co-planarity, Green: localization in BigCal and polar angle correlation with fit in Black
Blue: the background


8.54 GeV²
N
10^5
10^4
10^3
10^2
10^1
−0.05 −0.04 −0.03 −0.02 −0.01 −0.00 0.01 0.02 0.03 0.04 0.05
$(p_{HMS} - p_{\theta e})/p_{HMS}$
backgd 854 andrew 11/05/00

Preliminary Results from G_{Ep}(III)

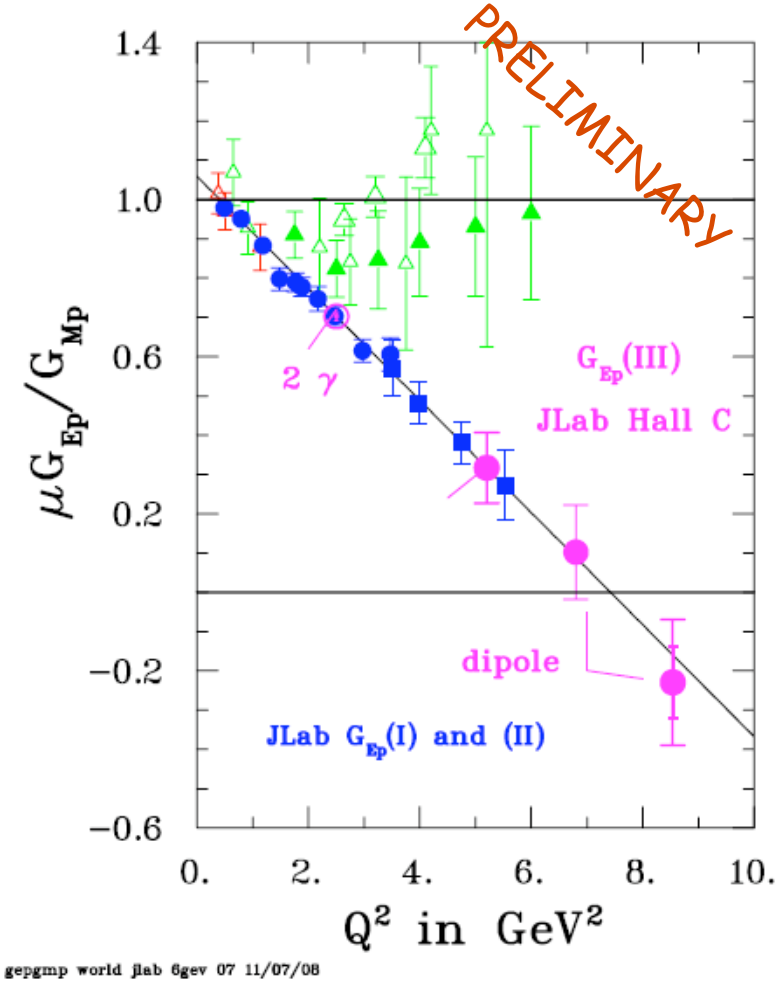
The preliminary results of G_{Ep}-III and G_{Ep}-2Gamma at JLab.

At 2.49 GeV^2, weighted average of 3 measurements at ε =0.14, 0.6[illegible] and 0.785. 0.01 statistics, very small point-to-point systematics

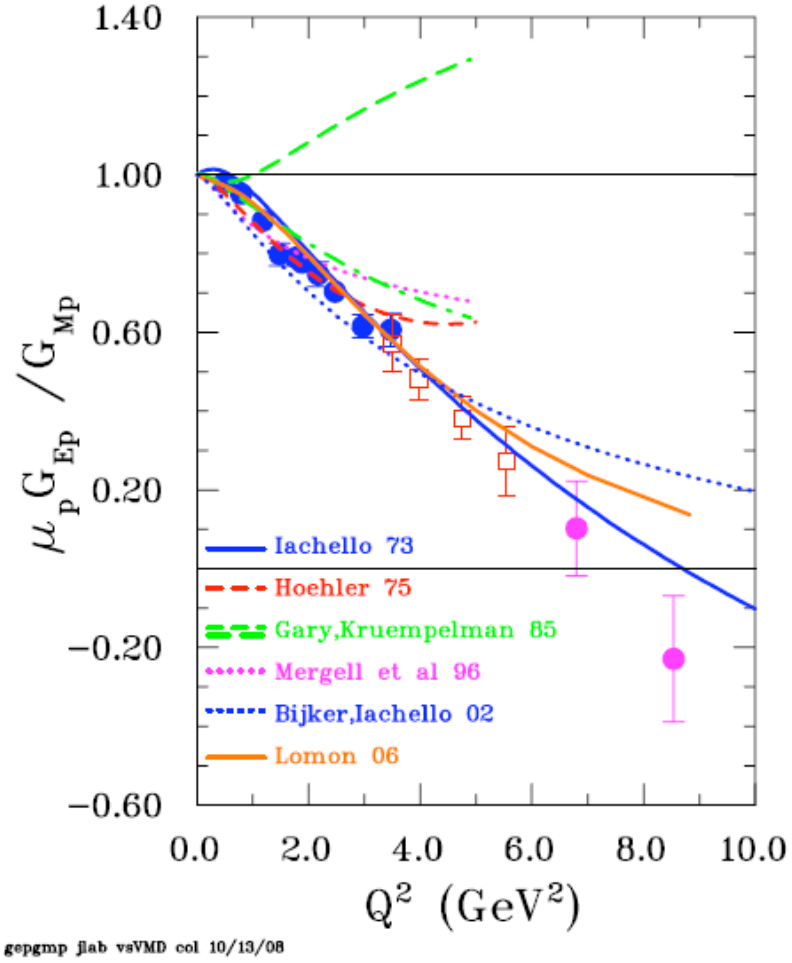
At 5.2 GeV^2, chosen for χ =180^0 spin precession, hence $P_n = P_\ell * \sin \chi \approx 0$;a check point.

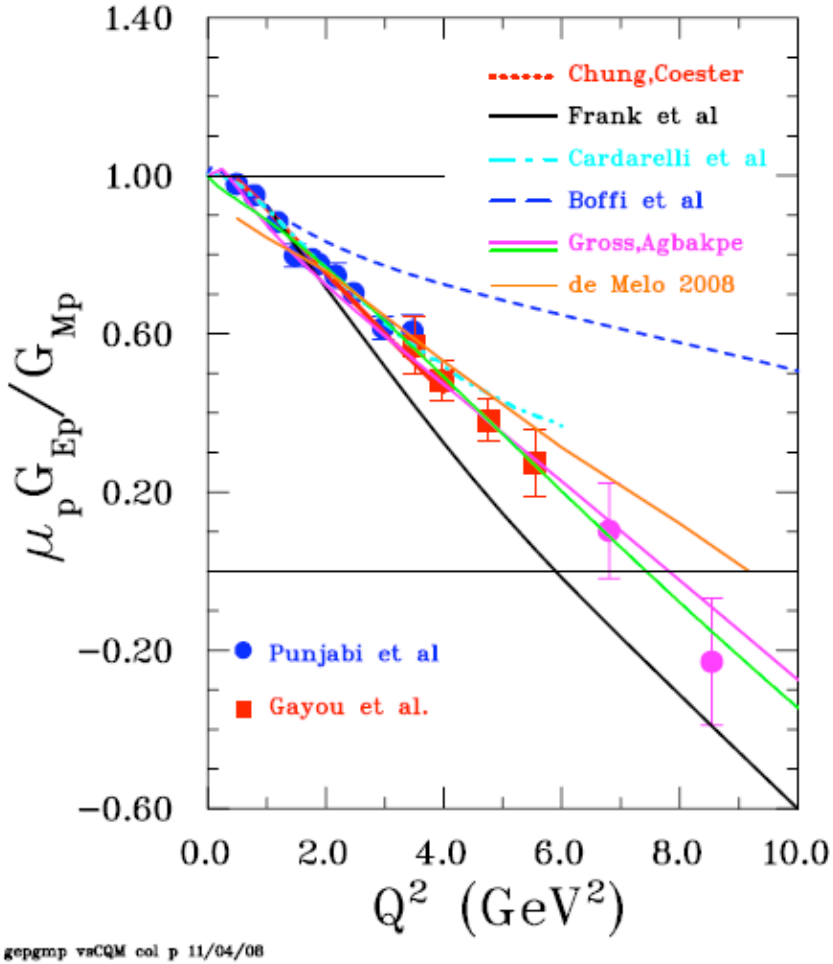
At 6.8 GeV^2, final error

At 8.54 GeV^2, current and proposal error bars shown; systematics not known yet.

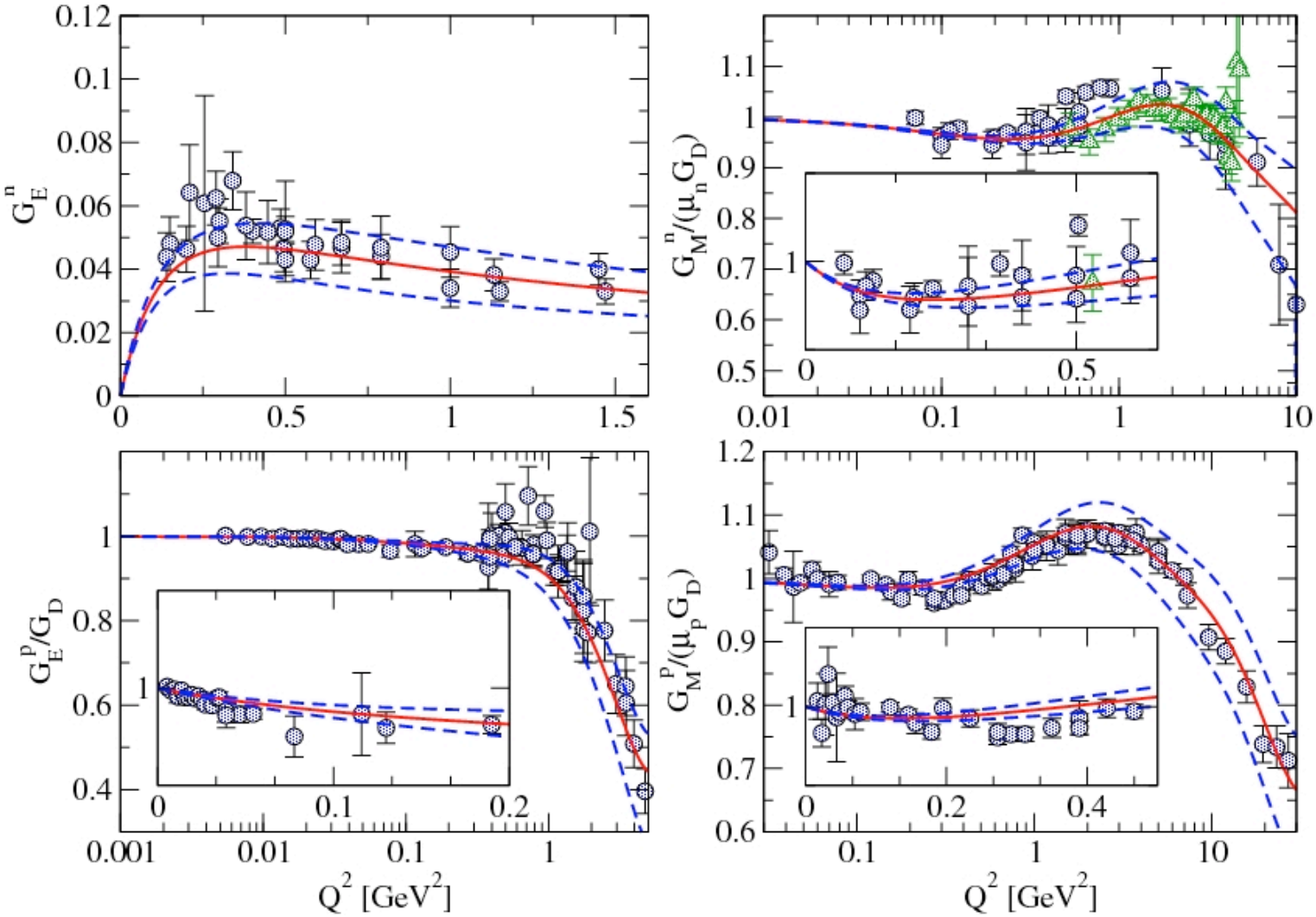


VMD and CQM


1.40
1.00
0.60
0.20
−0.20
−0.60
0.0
2.0
4.0
6.0
8.0
10.0
$\mu_p G_{Ep} / G_{Mp}$
Q^2 (GeV^2)
Iachello 73
Hoehler 75
Gary,Kruempelman 85
Mergell et al 96
Bijker,Iachello 02
Lomon 06
gepgmp jlab vsVMD col 10/13/08


1.40
1.00
0.60
0.20
−0.20
−0.60
0.0
2.0
4.0
6.0
8.0
10.0
$\mu_p G_{Ep} / G_{Mp}$
Q^2 (GeV^2)
Chung,Coester
Frank et al
Cardarelli et al
Boffi et al
Gross,Agbakpe
de Melo 2008
Punjabi et al
Gayou et al.
gepgmp vsCQM col p 11/04/08

Dispersion Relation Fits (Belushkin *et al.*)

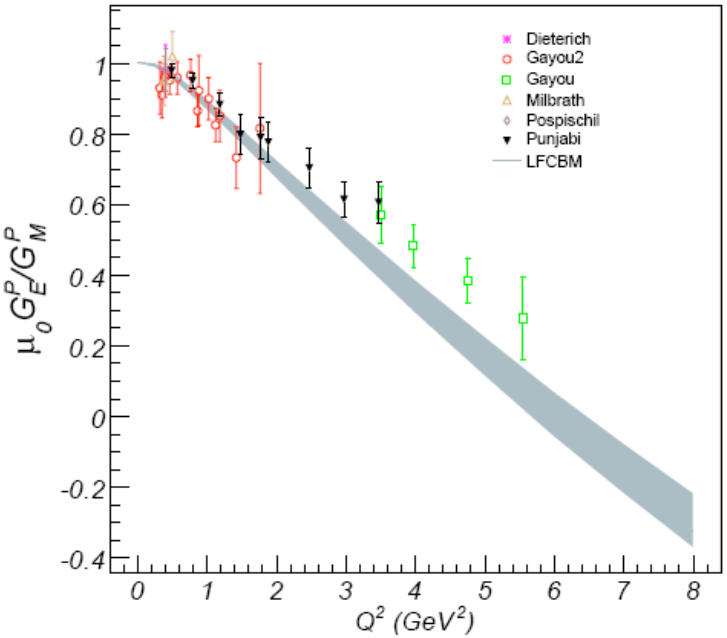


Lattice QCD, Fadeev

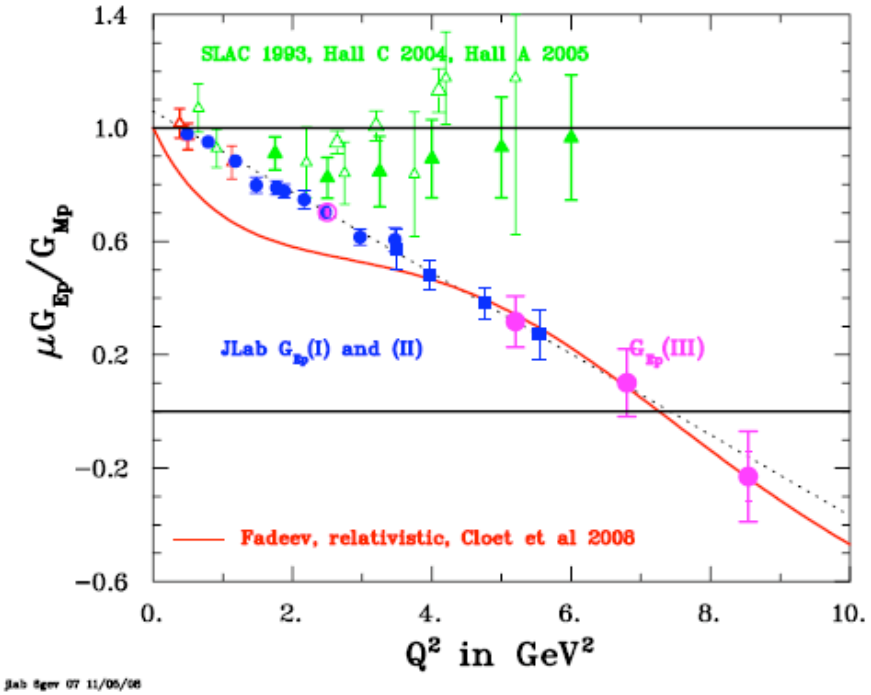
For example:

Matevosyan, Thomas,Miller(AIP 2005)
Use LFCBM (Light Front Cloudy Bag model, rCQM) to extrapolate QCDSF 'data' to physical pion mass, (Gloeckeler et al 2005)

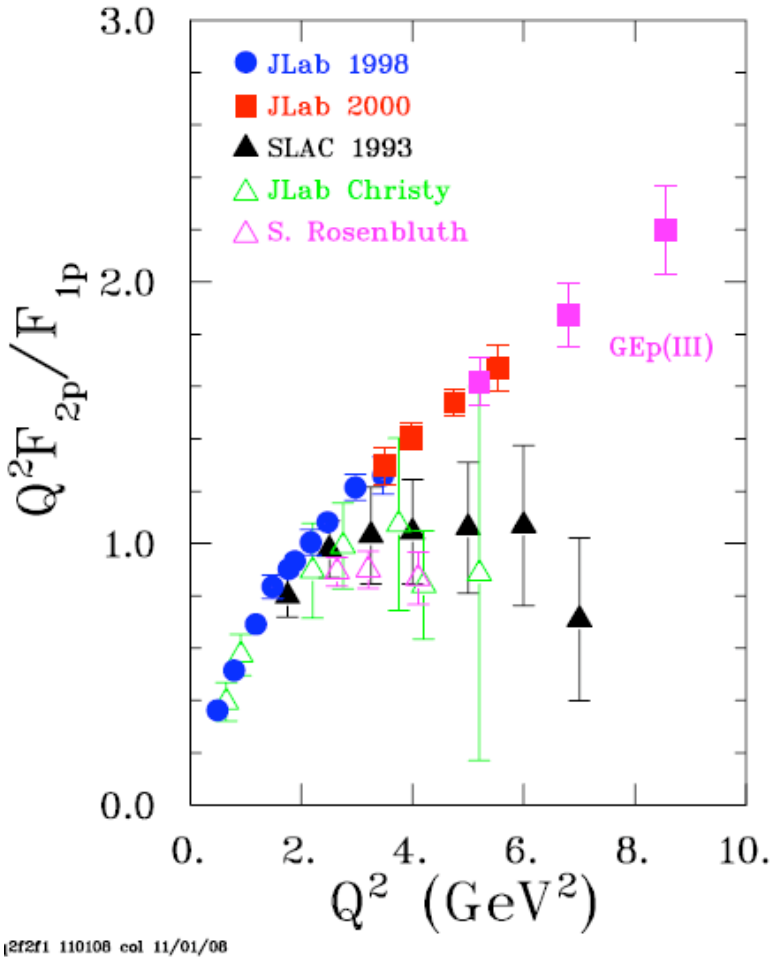
Poincare invariant Faddeev equation solution.
Arrington,C. Roberts and Zanetti et al.
J. Phys. G34, S23 (2007)
Cloet et al t b p 2008
Parameters: diquark magnetic dipole and quadrupole moments, scalar to vector EM transition strenath.

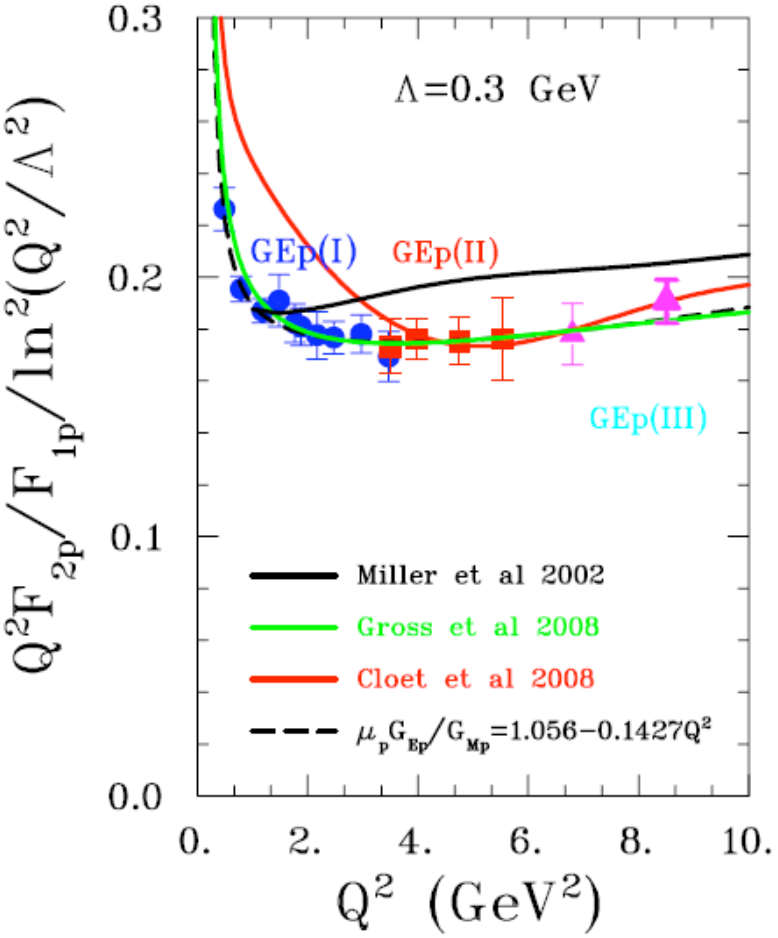


H.H. Matevosyan - PANIC 2005



Proton: F2 /F1 and pQCD





Brodsky and Farrar (75):
$Q^2F_2/F_1 \rightarrow$ constant

Belitsky, Ji and Yuan (03):
$Q^2F_2/F_1 \rightarrow \ln^2(Q^2/\Lambda^2)$

Conclusions

- Many competing/complementary theoretical models, with different approaches
- While most modern calculations describe the data well in the lower Q^2 regime, they begin to diverge significantly beyond the currently available data
- New data at higher Q^2 (for both proton and neutron) will place stringent constraints on available models, and will continue to motivate more advanced calculations

Gep-IV: Basic Plan

- Extend the measurement of G_{Ep}/G_{Mp} to the largest value of Q^2 possible using base equipment together with existing dectectors in Hall C
  - Hall C SHMS, equipped with the existing Hall C FPP, for proton detection
  - The existing BigCal detector for electron detector - perfect match to the SHMS
- Issues for Consideration:
  - Spin Precession in the SHMS
  - Radiation Damage to BigCal
  - CH_2 Analyzing Power

Gep-IV: Spin Precession in the SHMS

- The proton spin components precess in the magnetic elements of the SHMS, so that:

$$\begin{pmatrix} \mathbf{P_n^{fp}} \\ \mathbf{P_t^{fp}} \\ P_l^{fp} \end{pmatrix} = \begin{pmatrix} S_{nn} & \mathbf{S_{nt}} & \mathbf{S_{nl}} \\ S_{tn} & \mathbf{S_{tt}} & \mathbf{S_{tl}} \\ S_{ln} & S_{lt} & S_{ll} \end{pmatrix} \begin{pmatrix} P_n^{tar} \\ \mathbf{P_t^{tar}} \\ \mathbf{P_l^{tar}} \end{pmatrix}$$

- Because of the horizontal bender in the SHMS, the transverse and longitudinal components are significantly mixed:

$$\begin{aligned} P_t^{fp} &= P_t^{hb} = P_l^{tar} \sin\phi_{hb} + P_t^{tar} \cos\phi_{hb}, \\ P_n^{fp} &= P_l^{hb} \sin\phi_d = (P_l^{tar} \cos\phi_{hb} - P_t^{tar} \sin\phi_{hb}) \sin\phi_d \end{aligned}$$

Gep-IV: Spin Precession in the SHMS

- The uncertainties in the measured target polarizations are then given by:

$$(\Delta P_t^{tar})^2 = \frac{2}{NA^2}(\cos^2\phi_{hb} + \frac{\sin^2\phi_{hb}}{\sin^2\phi_d})$$

$$(\Delta P_l^{tar})^2 = \frac{2}{NA^2}(\sin^2\phi_{hb} + \frac{\cos^2\phi_{hb}}{\sin^2\phi_d})$$

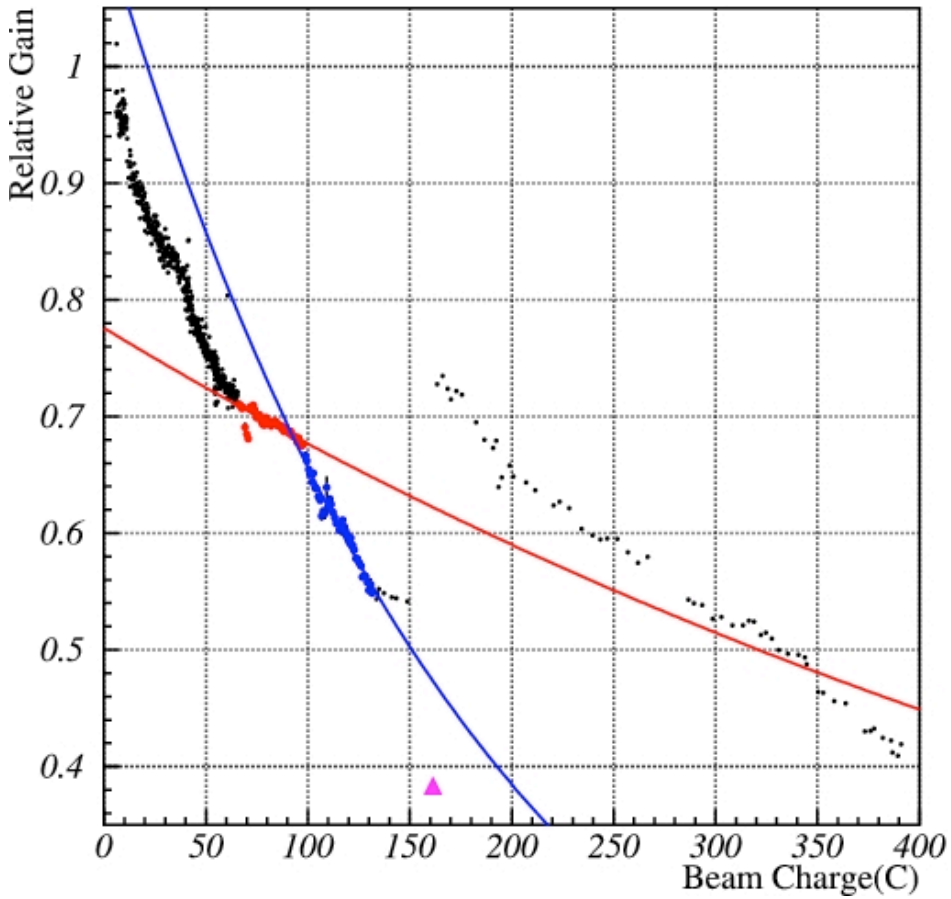
For $Q^2 = 13$ GeV^2, $\sin\phi_d \approx 1$, and thus $\cos^2\phi_{hb} + \frac{\sin^2\phi_{hb}}{\sin^2\phi_d} \approx 1$

Gep-IV:BigCal Radiation Damage

- The relative light output in the BigCal lead glass will decrease significantly with accumulated beam charge.
- This does affect energy resolution in BigCal, but we are fairly insensitive to this.
- The main concern is being able to set a relatively high hardware threshold to keep the BigCal rates low.

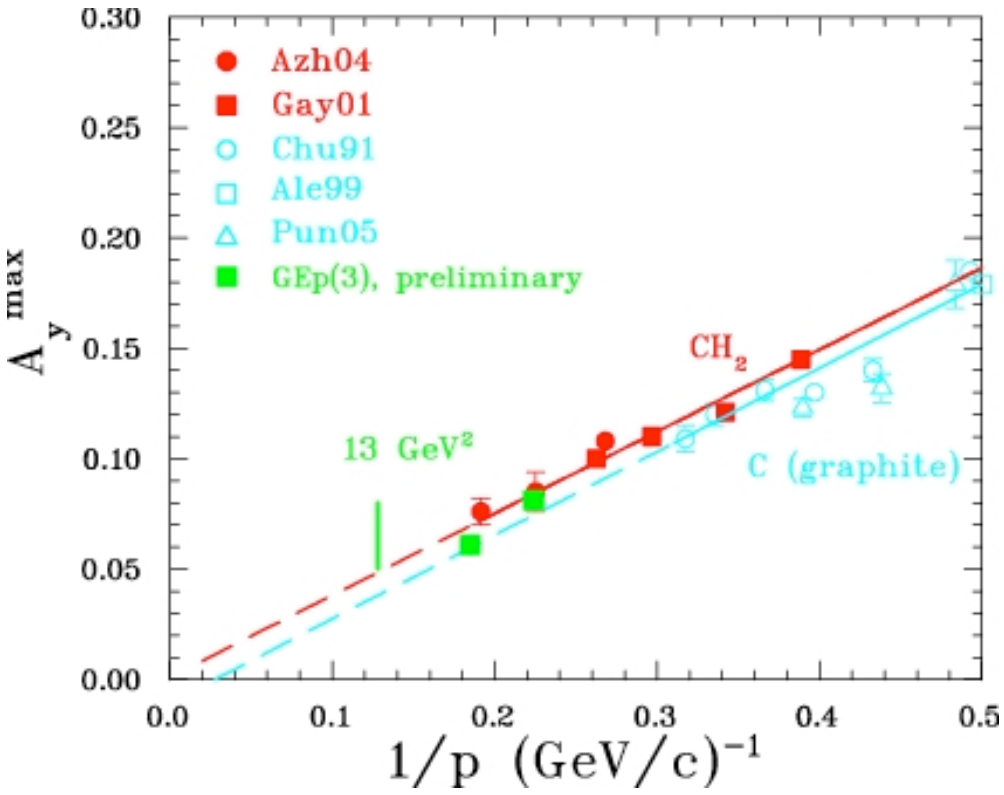
Gep-IV:BigCal Radiation Damage

- Data from Gep-III was used to estimate gain loss rate in Gep-IV
- GEANT simulation of the energy flux per Coulomb through the front face of the calorimeter
- Result: Curing about once per week in Gep-IV
- Use maintenance days - need four hours of curing to recover one week of damage



Gep-IV: CH_2 Analyzing Power

- As a by-product of the polarization transfer experiments, we can extract the (average/maximum) CH_2 analyzing power in the FPP
- Empirically, the maximum analyzing power scales as 1/p, and the shape of the distribution is largely momentum independent; this allows us to make accurate predictions of the analyzing power at various momenta
- In addition, full GEANT3 simulations have been performed to estimate the scattering efficiency (describes Gep-III data well)



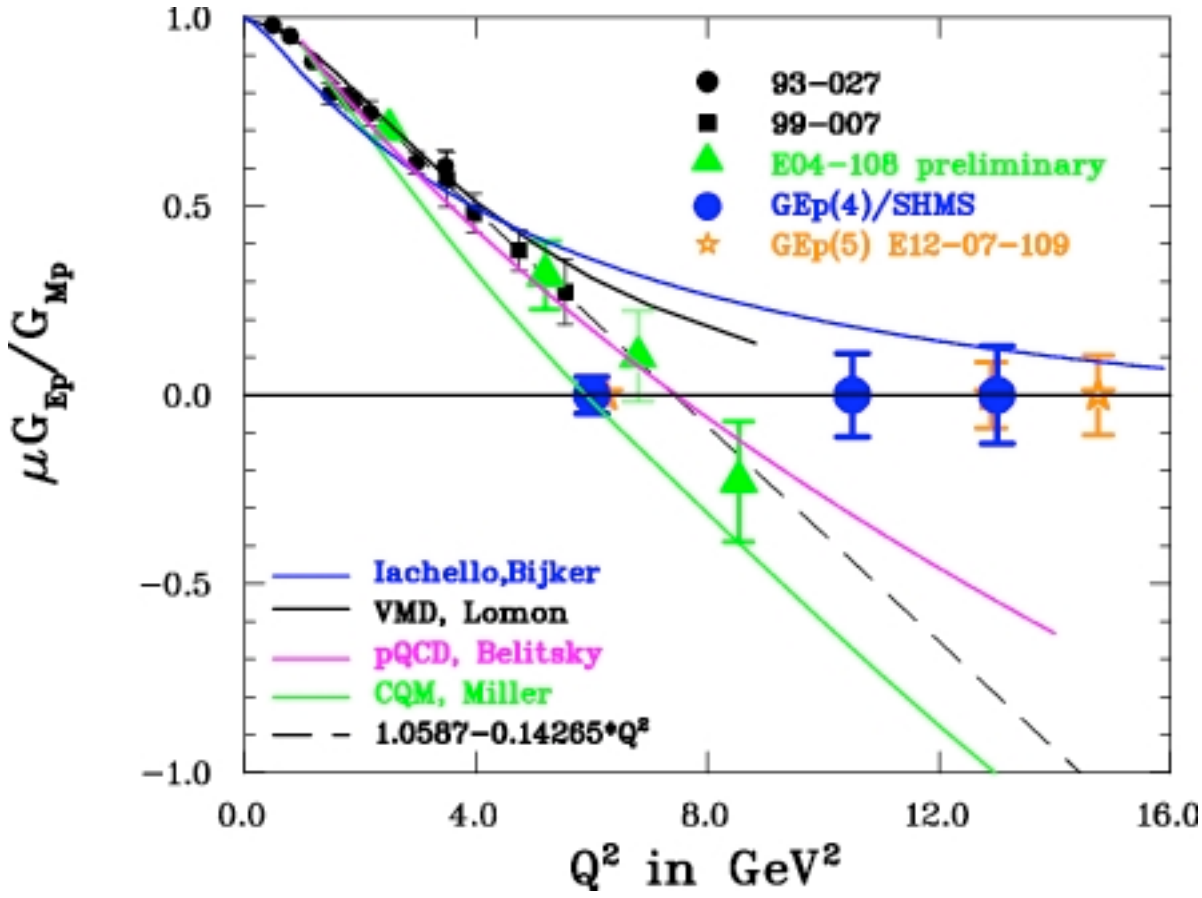
Gep-IV: Kinematics

Q^2	E_e	θ_e	$E_{e'}$	θ_p	p_p	$d\sigma/d\Omega_e$	ϵ	χ	$\Delta\Omega_e$
GeV^2	GeV	deg	GeV	deg	GeV/c	cm^2/sr		deg	msr
6	6.6	30	3.4	25	4.03	1.1×10^{-35}	0.72	145.4	8.6
10.5	8.8	35.5	3.20	16.7	6.47	3.5×10^{-37}	0.55	229.7	24
13	11.0	31.3	4.07	15.7	7.81	1.6×10^{-37}	0.58	276.5	23

Table 2: *The proposed kinematics. Assumed SHMS spectrometer solid angle: 5 msr. Assumed beam characteristics: 75 μA, 85% polarization. Assumed target: 30 cm LH_2.*

Q^2	E_e	COM	absolute $\Delta(G_{Ep}/G_{Mp})^*$	time
GeV^2	GeV			days
6.0	**6.6**	**3.9×10^{-3}**	**0.04**	**4**
10.5	**8.8**	**1.5×10^{-3}**	**0.11**	**30**
13.0	**11.0**	**1.1×10^{-3}**	**0.13**	**60**

Gep-IV: Predictions



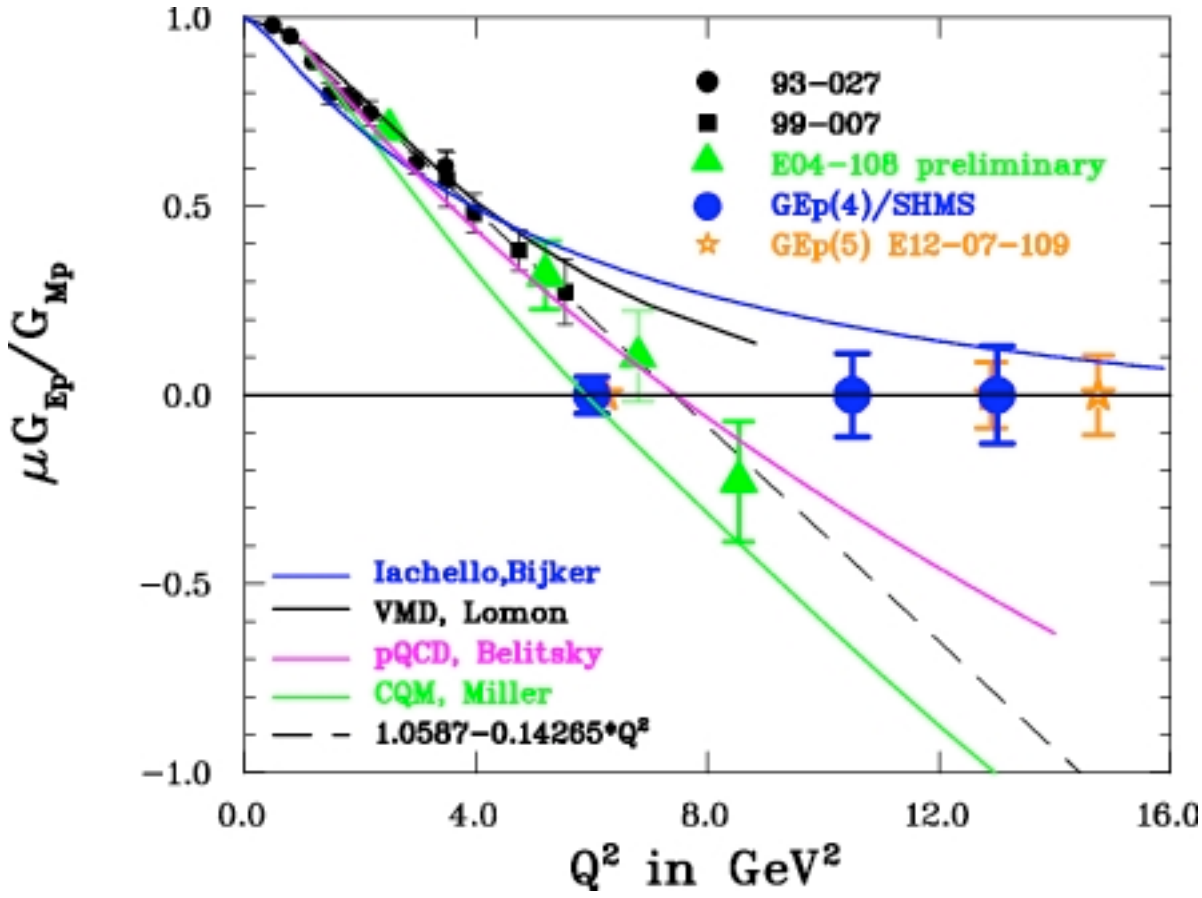
Gep-IV vs. Gep-V

- Advantages of Gep-IV:
  - Cleaner selection of elastic events - data from Gep-III show that polarization of inelastic background is very different than elastic data (not surprising)
  - Gep-IV uses either existing or base Hall C equipment - this has technical as well as financial/funding advantages
  - Gep-IV can be run very early post-upgrade
  - Software algorithms are refined and well-tested
- It is always an advantage to have more than one measurement with different systematics (see Rosenbluth data!)
- At this point in time, it makes good sense to have both of these experiments on the books, given the various uncertainties (technical and funding) surrounding the Hall A experiment

Conclusions

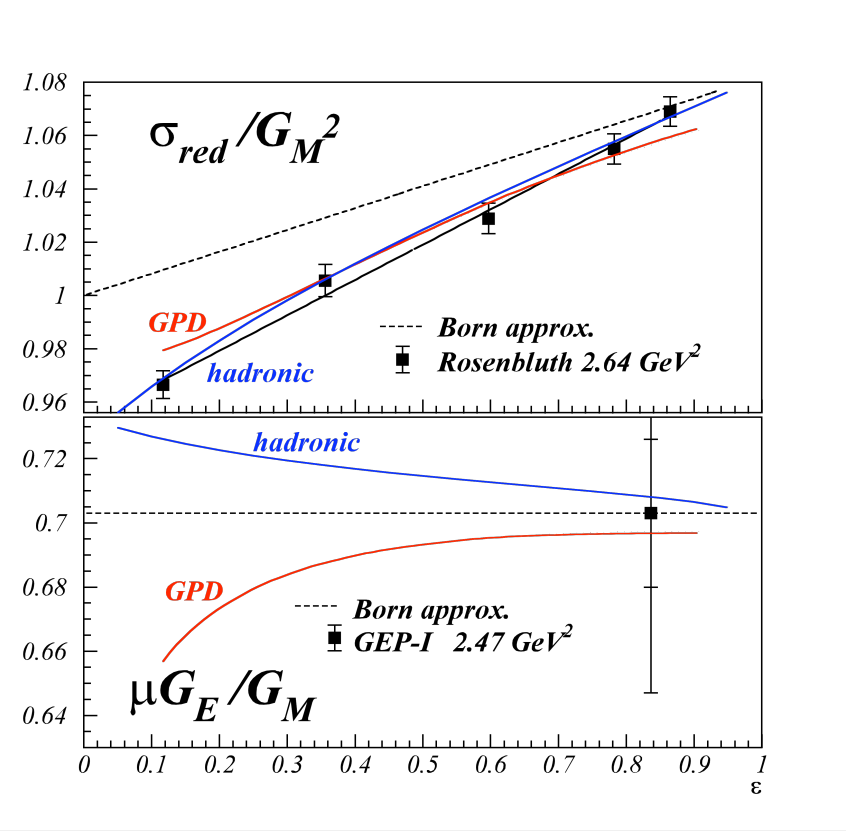
- Many competing/complementary theoretical models, with different approaches
- While most modern calculations describe the data well in the lower Q^2 regime, they begin to diverge significantly beyond the currently available data
- New data at higher Q^2 (for both proton and neutron) will place stringent constraints on available models, and will continue to motivate more advanced calculations
- Gep-IV will provide high quality data on the form factor ratio up to Q^2 = 13 GeV^2, using existing/base equipment. The experiment can be carried out as soon as the 11 GeV electron beam is available in Hall C - could even be used as a commissioning experiment for the SHMS
- No major technical issues face this experiment - hardware and software are "ready to go"

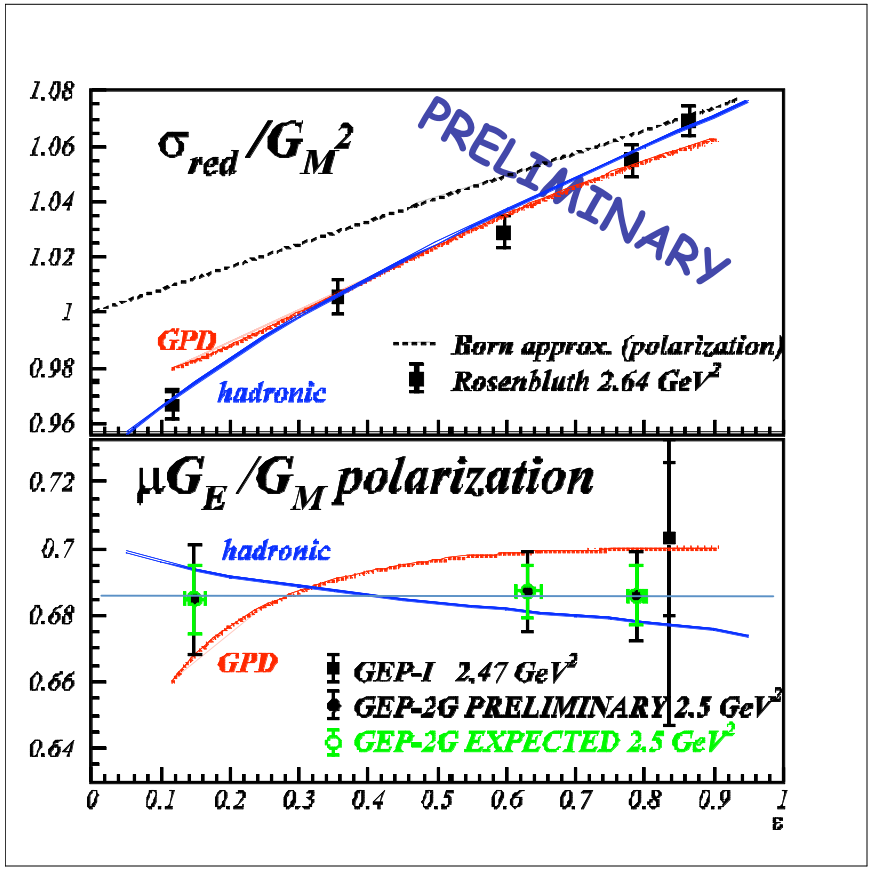
Gep-IV: Predictions



Backup Slides

2 γ -Gamma Models and Data

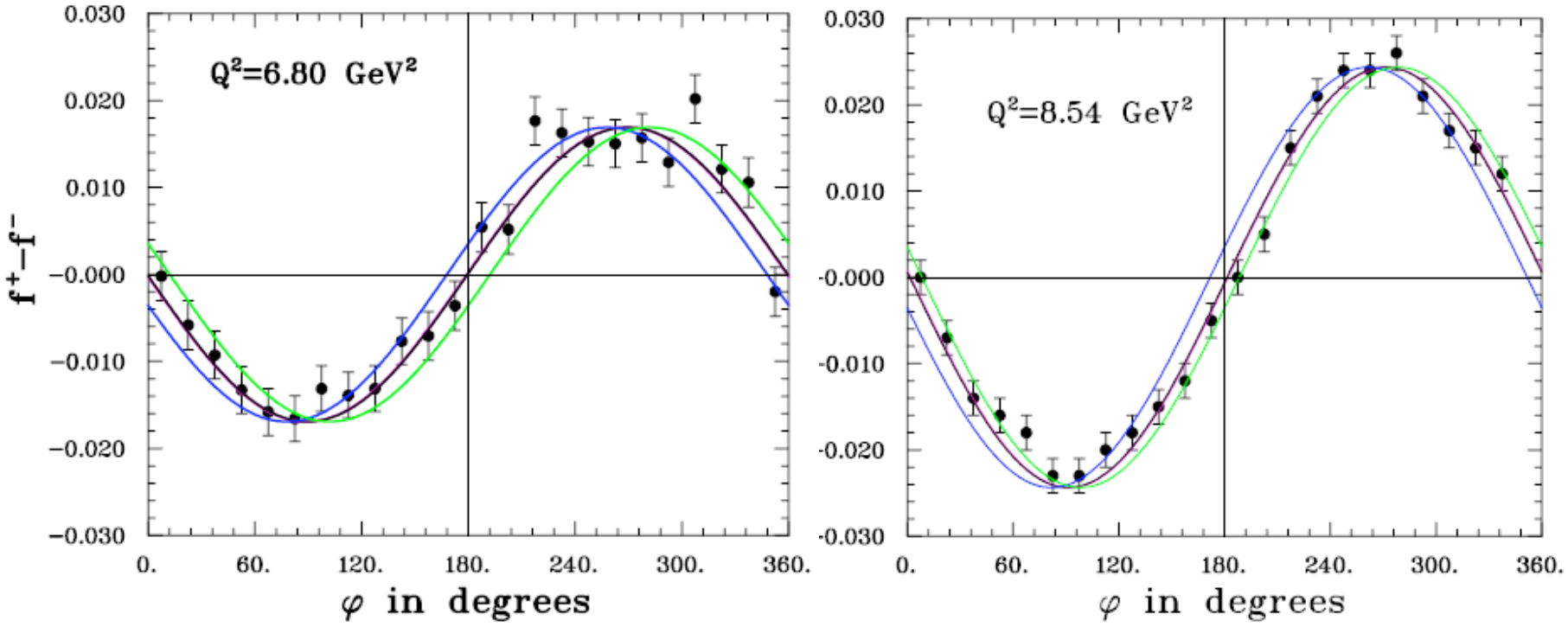




The inclusion of hard 2 γ exchange:
Chen et al (2003) with GPDs,
Blunden et al (2003) in hadronic model
Several other calculations exist!
Predictions range between no effect and approximative reproduction of data.

Expt. Jlab 04-019 measured Ratio at 3 values of ε for Q^2=2.49 GeV^2; no ε-dependence at 0.01 level

Physical Asymmetries at Q^2 of 6.8 and 8.5 GeV2



Constituent Quark Models (Understanding the Dynamics)

Initially proposed by Isgur & Karl (78)
Early non-relativistic CQM

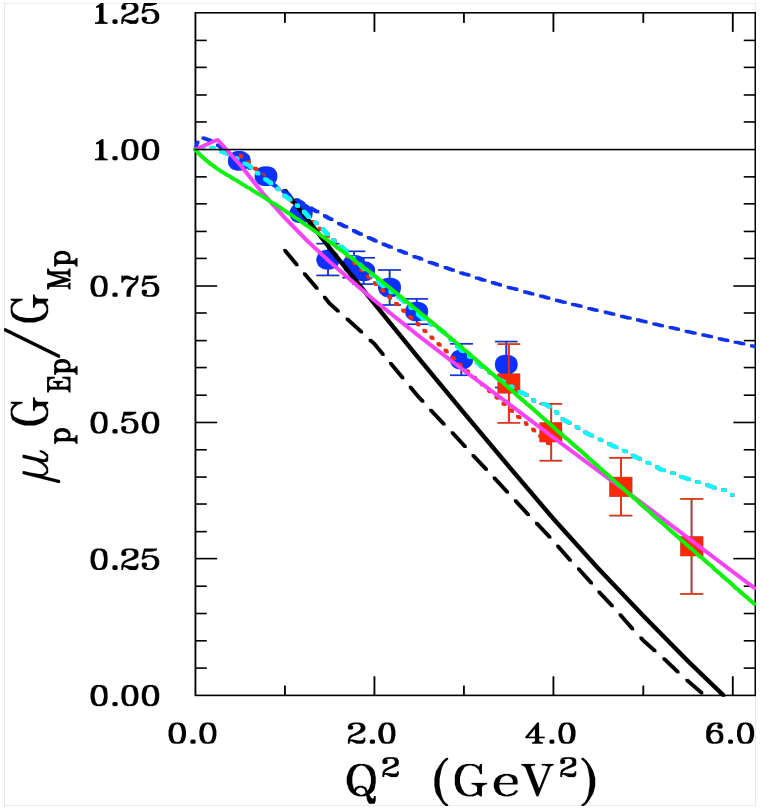
Relativistic CQM: 3 forms of dynamics: instant, point like and light-front

Point-like: Chung & Coester (91) .....
Cardarelli et al (00) ——

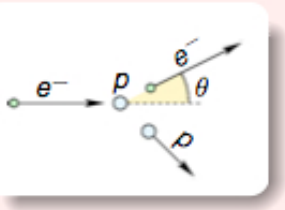
Front form:
Frank,Jennings, Miller (96) – – – –
G.A. Miller, Frank (02) ——

Goldstone boson: Boffi et al (01) – · –

Cov. spectator: Gross,Agbakpe (04) – – –
Gross,Ramalho,Pena (06) ——

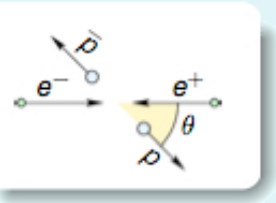


Conclusion: Relativistic CQ are relevant degrees of freedom



Elastic scattering

$$\frac{d\sigma}{d\Omega} = \frac{\alpha^2 E'_e \cos^2\frac{\theta}{2}}{4E_e^3 \sin^4\frac{\theta}{2}} \left[G_E^2 + \tau \left(1 + 2(1+\tau)\tan^2\frac{\theta}{2} \right) G_M^2 \right] \frac{1}{1+\tau}$$

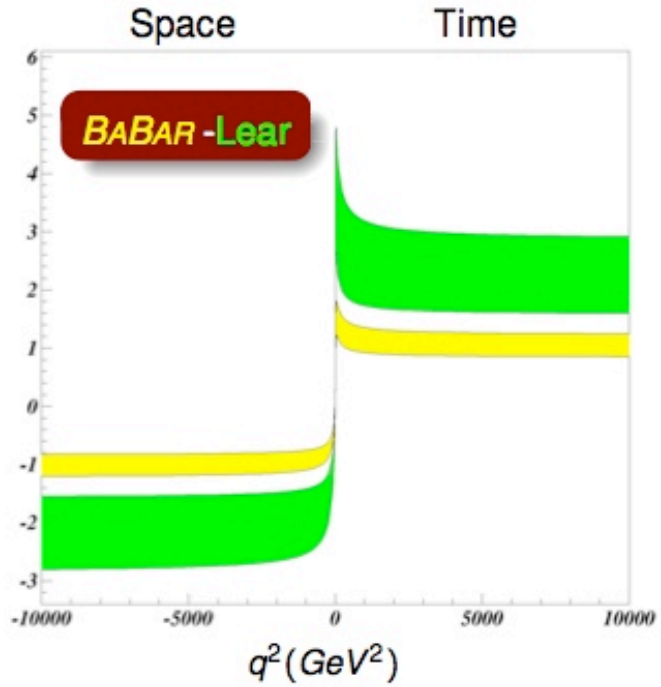


Annihilation

$$\frac{d\sigma}{d\Omega} = \frac{\alpha^2 \sqrt{1-1/\tau}}{4q^2} C \left[(1+\cos^2\theta)|G_M|^2 + \frac{1}{\tau}\sin^2\theta |G_E|^2 \right]$$

Coulomb correction: $C \approx \frac{y}{1-e^y}$ $\qquad y = \frac{\pi\alpha M_p}{\beta q}$

Asymptotic behaviour of $G_E^p(q^2)/G_M^p(q^2)$



pQCD prediction

$$\left| \frac{G_E^p(q^2)}{G_M^p(q^2)} \right| \xrightarrow[|q^2|\to\infty]{} 1$$

Rinaldo Baldini
Jlab Exclusive Workshop 2007